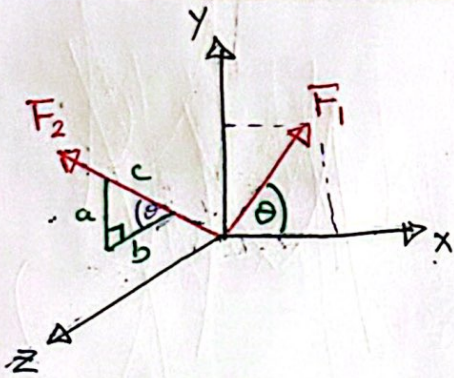


* Resolution of a Force in (3D)

(Jul 25)

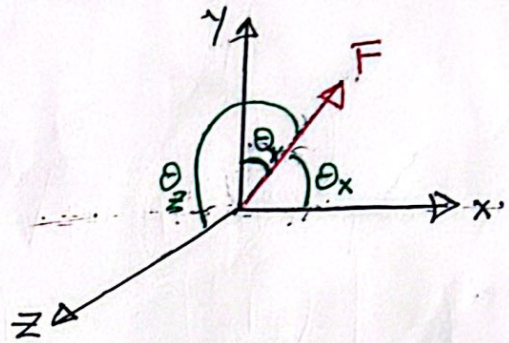
1 One Angle or Triangle



$$\vec{F}_1 = (F_1 \cos \theta) \hat{i} + (F_1 \sin \theta) \hat{j}$$

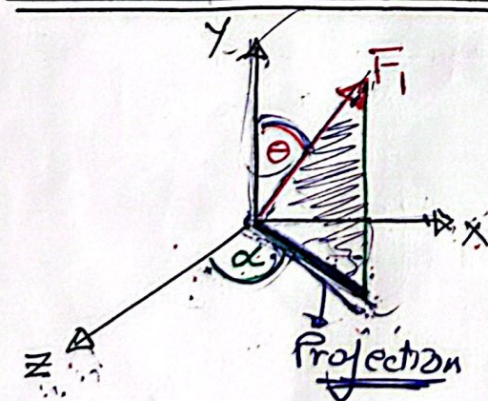
$$\vec{F}_2 = (F_2 \cdot \frac{a}{c}) \hat{j} + (F_2 \cdot \frac{b}{c}) \hat{k}$$

2 Three Angles



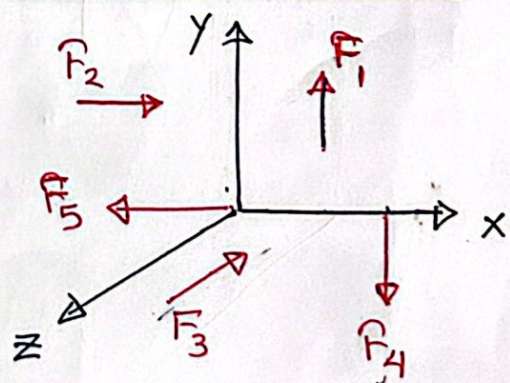
$$\vec{F} = (F \cos \theta_x) \hat{i} + (F \cos \theta_y) \hat{j} + (F \cos \theta_z) \hat{k}$$

3 One Angle and Projection + angle



$$\vec{F}_1 = (F_1 \sin \theta \cdot \sin \alpha) \hat{i} + (F_1 \cos \theta) \hat{j} + (F_1 \sin \theta \cdot \cos \alpha) \hat{k}$$

4 Force // axis



$$\begin{aligned} \vec{F}_1 &= (F_1) \hat{j} & \vec{F}_3 &= -(F_3) \hat{k} \\ \vec{F}_2 &= (F_2) \hat{i} & \vec{F}_4 &= -(F_4) \hat{j} \\ \vec{F}_5 &= -(F_5) \hat{i} \end{aligned}$$

5 Points

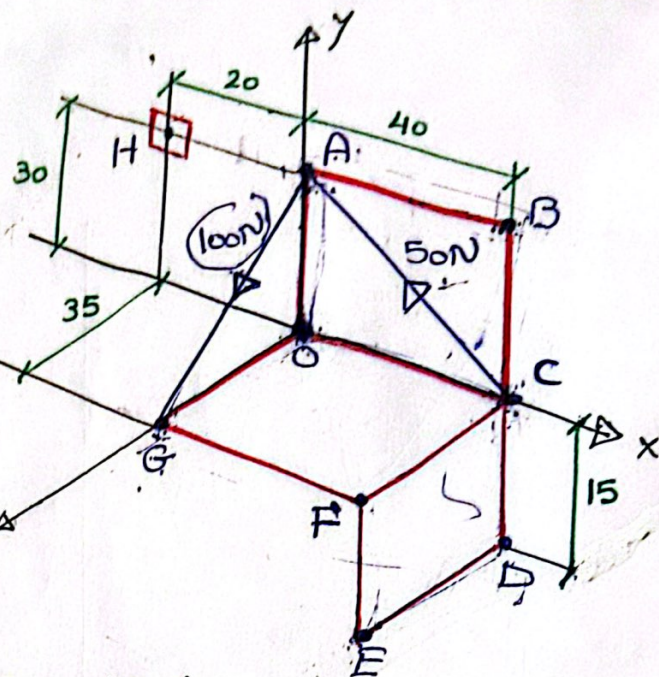
(x, y, z)

A (0, 30, 0) E (40, -15, 35)

B (40, 30, 0) F (40, 0, 35)

C (40, 0, 0) G (0, 0, 35)

D (40, -15, 0) H (-20, 30, 0)



Force = (magnitude of force) * (λ)

$\vec{F}_{AG} = (F_{AG}) * \lambda_{AG}$

Where

$\lambda_{AG} = \frac{\vec{AG}}{|AG|}$

$\vec{AG} = G - A = 0i - 30j + 35k$

$|AG| = \sqrt{0^2 + 30^2 + 35^2} = 46$

$\therefore \vec{F}_{AG} = (100) * \left[\frac{-30j + 35k}{46} \right] = -65.2j + 76k$

$\vec{F}_{CA} = (F_{CA}) * \lambda_{CA}$

$= (50) * \left[\frac{-40i + 30j + 0k}{\sqrt{40^2 + 30^2 + 0^2}} \right] = -40i + 30j$

$\frac{50 * -40}{\sqrt{40^2 + 30^2}}$

$\frac{50 * 30}{\sqrt{40^2 + 30^2}}$

2

* Resultant of Forces in (3D) (Example)

$$\vec{F}_1 = (a)\hat{i} + (b)\hat{j} + (c)\hat{k}$$

$$\vec{F}_2 = (d)\hat{i} + (e)\hat{j} + (f)\hat{k}$$

$$\therefore \text{Resultant } (R) = \vec{F}_1 + \vec{F}_2$$

$$R = \underbrace{(a+d)}_{\rightarrow R_x} \hat{i} + \underbrace{(b+e)}_{\rightarrow R_y} \hat{j} + \underbrace{(c+f)}_{\rightarrow R_z} \hat{k}$$

$$\rightarrow \text{magnitude of Resultant } |R| = \sqrt{R_x^2 + R_y^2 + R_z^2} = \checkmark$$

$\rightarrow$ direction of Resultant :

$$\theta_x = \cos^{-1} \left(\frac{R_x}{|R|} \right) \quad \theta_y = \cos^{-1} \left(\frac{R_y}{|R|} \right) \quad \theta_z = \cos^{-1} \left(\frac{R_z}{|R|} \right)$$

* Equilibrium of a Point in (3D)

$$\Sigma F_x = 0$$

(i) مجموع

$$\Sigma F_y = 0$$

(j) مجموع

$$\Sigma F_z = 0$$

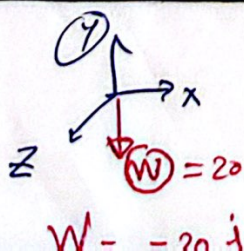
(k) مجموع

mode $\rightarrow 5 \rightarrow 2$

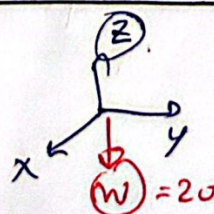
Note : $\text{Weight} = \text{mass} \times \text{gravity}$

$\downarrow$ $\downarrow$ $\downarrow$
Newton kg 9.81 m/s²

(↓) اتجاه (weight) رأسى لـ (je)



3



$$W = -20\hat{k}$$

Determine the tension in cables BA and BC necessary to support the 60-kg cylinder in the figure shown.

$$m = 60 \text{ kg}, W = 60 \times 9.81 = 588.6 \text{ N}$$

$$\vec{T}_{BC} = (T_{BC} \cos 45^\circ) \hat{i} + (T_{BC} \sin 45^\circ) \hat{j}$$

$$\vec{T}_{BA} = -(T_{BA} \cdot \frac{4}{5}) \hat{i} + (T_{BA} \cdot \frac{3}{5}) \hat{j}$$

$$\vec{W} = -588.6 \hat{j}$$

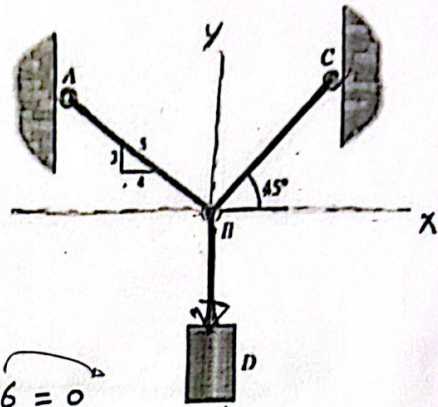
$$\sum F_x = 0: T_{BC} \cos 45^\circ - T_{BA} \cdot \frac{4}{5} = 0$$

$$\sum F_y = 0: T_{BC} \sin 45^\circ + T_{BA} \cdot \frac{3}{5} - 588.6 = 0$$

$$T_{BC} = 475.6 \text{ N}$$

$$T_{BA} = 420.4 \text{ N}$$

mode $\rightarrow S \rightarrow 1$



The spring has a stiffness of 800 N/m and an unstretched length of 200 mm. Determine the force in cables BC and BD when the spring is held in position.

$$K = 800 \text{ N/m} \quad l_0 = 200 \text{ mm}$$

$$= 0.8 \text{ N/mm} \quad l = 500 \text{ mm}$$

$$F_{sp} = K(\Delta x)$$

$$= (500 - 200) = 240 \text{ N}$$

$$\vec{F}_{sp} = -240 \hat{i}$$

$$\vec{T}_{BC} = (T_{BC} \cdot \frac{400}{565.6}) \hat{i} + (T_{BC} \cdot \frac{400}{565.6}) \hat{j}$$

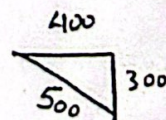
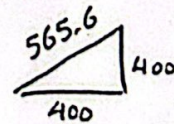
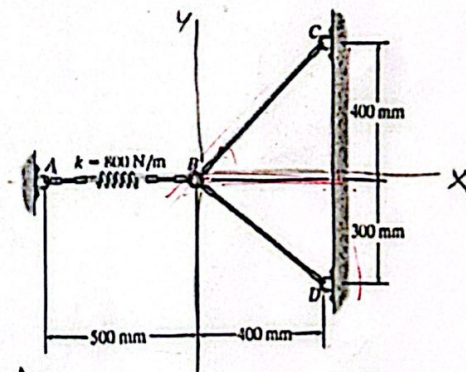
$$\vec{T}_{BD} = (T_{BD} \cdot \frac{400}{500}) \hat{i} - (T_{BD} \cdot \frac{300}{500}) \hat{j}$$

$$\sum F_x = 0: T_{BC} \cdot \frac{400}{565.6} + T_{BD} \cdot \frac{400}{500} - 240 = 0$$

$$\sum F_y = 0: T_{BC} \cdot \frac{400}{565.6} - T_{BD} \cdot \frac{300}{500} = 0$$

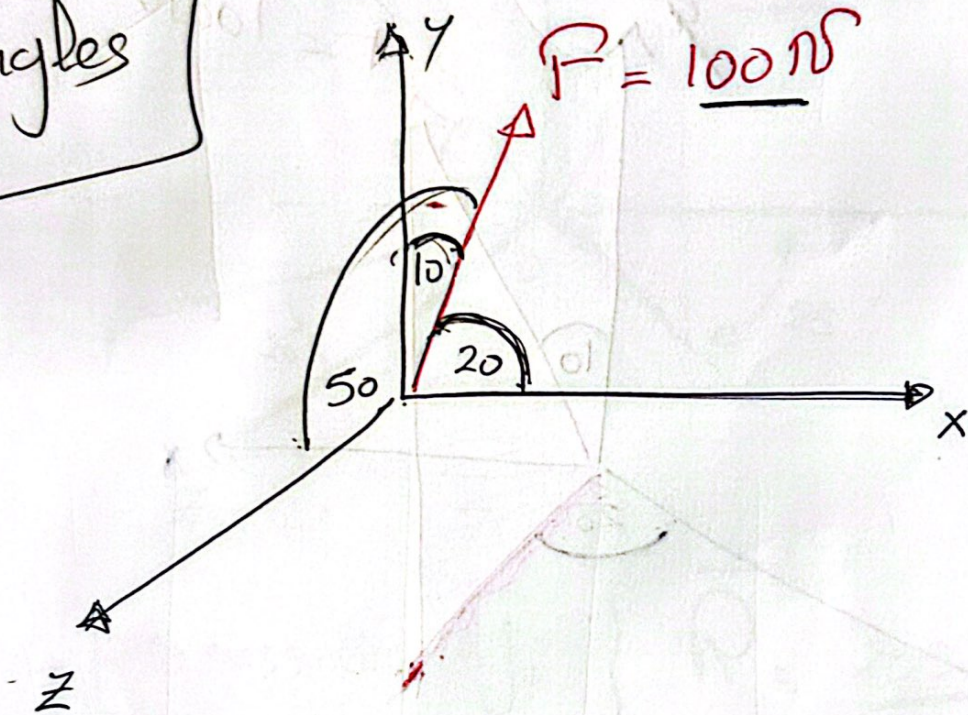
$$T_{BC} = 145.4 \text{ N}$$

$$T_{BD} = 171.4 \text{ N}$$



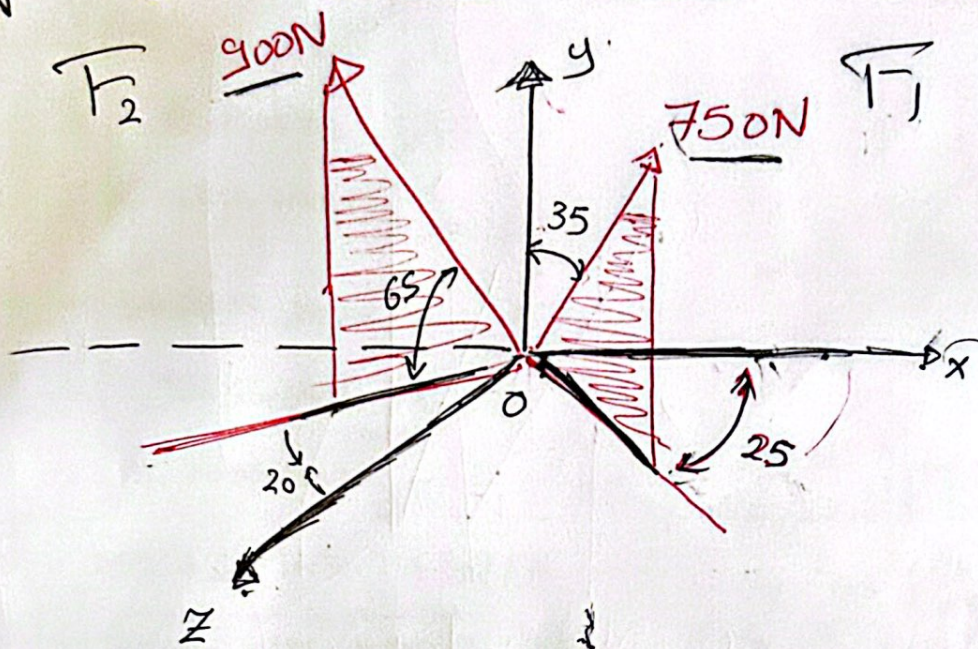
Resolution in 3D

3 angles



$$\begin{aligned} \vec{F} = & + [100 \cos 20] \hat{i} \\ & + [100 \cos 10] \hat{j} \\ & + [100 \cos 50] \hat{k} \end{aligned}$$

Find magnitude / direction of Resultant



$$\vec{F}_1 = + [750 \sin 35 \cdot \cos 25] \hat{i} \\ + [750 \cos 35] \hat{j} \\ + [750 \sin 35 \cdot \sin 25] \hat{k}$$

$$\vec{F}_2 = - [900 \cos 65 \cdot \sin 20] \hat{i} \\ + [900 \sin 65] \hat{j} \\ + [900 \cos 65 \cdot \cos 20] \hat{k}$$

$$\vec{R} = 299.7 \hat{i} \\ + 1430 \hat{j} \\ + 527.1 \hat{k}$$

magnitude $|\vec{R}|$

$$= \sqrt{299.7^2 + 1430^2 + 527.1^2} \\ = 1550.1 \text{ N}$$

direction:

$$\theta_x = \cos^{-1} \left(\frac{299.7}{1550.1} \right) = 80.3^\circ$$

$$\theta_y = \cos^{-1} \left(\frac{1430}{1550.1} \right) = 22.7^\circ$$

$$\theta_z = \cos^{-1} \left(\frac{527.1}{1550.1} \right) = 69.6^\circ$$

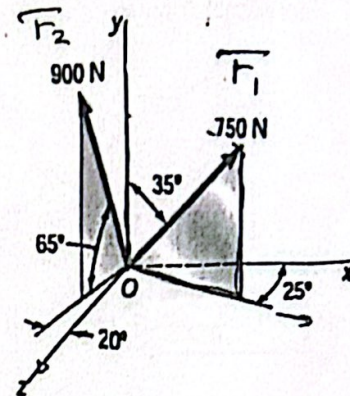
2. Find the magnitude and coordinate direction angles of the resultant force.

$$\vec{F}_1 = +[750 \sin 35^\circ \cos 25^\circ] \hat{i} \\ + [750 \cos 35^\circ] \hat{j} \\ + [750 \sin 35^\circ \sin 25^\circ] \hat{k}$$

$$\vec{F}_2 = -[900 \cos 65^\circ \sin 20^\circ] \hat{i} \\ + [900 \sin 65^\circ] \hat{j} \\ + [900 \cos 65^\circ \cos 20^\circ] \hat{k}$$

$$\vec{R} = 259.7 \hat{i} + 1430 \hat{j} + 527.1 \hat{k}$$

$$|\vec{R}| = \sqrt{259.7^2 + 1430^2 + 527.1^2} = 1550.1$$



direction:

$$\theta_x = \cos^{-1} \left(\frac{259.7}{1550.1} \right) = 80.3^\circ$$

$$\theta_y = \cos^{-1} \left(\frac{1430}{1550.1} \right) = 22.7^\circ$$

$$\theta_z = \cos^{-1} \left(\frac{527.1}{1550.1} \right) = 69.6^\circ$$

Determine the magnitude and coordinate direction angles of the resultant force.

$$\vec{F}_1 = +[400 \cos 45^\circ] \hat{i} + [400 \cos 60^\circ] \hat{j} \\ - [400 \cos 60^\circ] \hat{k}$$

$$\vec{F}_2 = +[125 \times \frac{4}{5} \cos 20^\circ] \hat{i} \\ - [125 \times \frac{4}{5} \sin 20^\circ] \hat{j} \\ + [125 \times \frac{3}{5}] \hat{k}$$

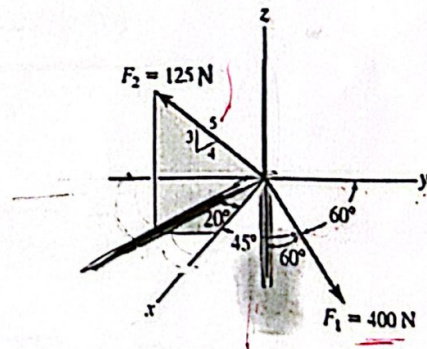
$$\vec{R} = 376.8 \hat{i} + 165.79 \hat{j} - 125 \hat{k}$$

magnitude: $|\vec{R}| = \sqrt{376.8^2 + 165.79^2 + 125^2} = 430.2 \text{ N}$

direction: $\theta_x = \cos^{-1} \left(\frac{376.8}{430.2} \right) = 28.85^\circ$

$$\theta_y = \cos^{-1} \left(\frac{165.79}{430.2} \right) = 67.3^\circ$$

$$\theta_z = \cos^{-1} \left(\frac{-125}{430.2} \right) = 106.8^\circ$$



★★

5. A 200-kg cylinder is hung by means of two cables AB and AC, which are attached to the top of a vertical wall. A horizontal force P perpendicular to the wall holds the cylinder in the position shown. Determine the magnitude of P and the tension in each cable.

$$m = 200 \text{ kg}$$

$$W = 200 \times 9.81 = 1962 \text{ N}$$

$$\vec{W} = -1962 \hat{j}$$

$$\vec{T}_{AB} = (T_{AB}) \left(\frac{-1.2\hat{i} + 10\hat{j} + 8\hat{k}}{\sqrt{1.2^2 + 10^2 + 8^2}} \right)$$

$$= (T_{AB}) (-0.093\hat{i} + 0.77\hat{j} + 0.62\hat{k})$$

$$\vec{T}_{AC} = (T_{AC}) \left(\frac{-1.2\hat{i} + 10\hat{j} - 10\hat{k}}{\sqrt{1.2^2 + 10^2 + 10^2}} \right)$$

$$= (T_{AC}) (-0.084\hat{i} + 0.7\hat{j} - 0.7\hat{k})$$

$$\vec{P} = (P)\hat{i}$$

$$\sum F_x = 0: -0.093 T_{AB} - 0.084 T_{AC} + P = 0$$

$$\sum F_y = 0: 0.77 T_{AB} + 0.7 T_{AC} - 1962 = 0$$

$$\sum F_z = 0: 0.62 T_{AB} - 0.7 T_{AC} = 0$$

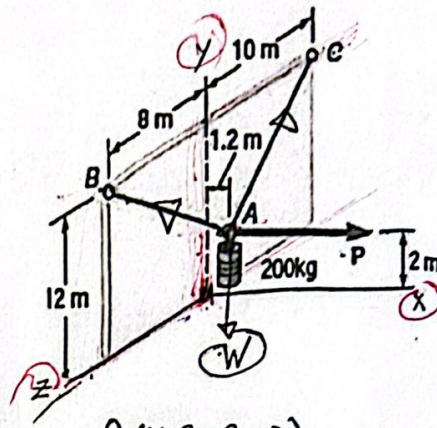
$$T_{AB} = 1411.5 \text{ N}$$

$$T_{AC} = 1250.1 \text{ N}$$

$$P = 247.5 \text{ N}$$

mode → 5 → 2

mode → 9 → 1 → 3



$$A(1.2, 2, 0)$$

$$B(0, 12, 8)$$

$$C(0, 12, -10)$$

$$\underline{\sum F_x = 0} : P - 0.093 T_{AB} - 0.084 T_{AC} = 0$$

$$\underline{\sum F_y = 0} : -1962 + 0.77 T_{AB} + 0.7 T_{AC} = 0$$

$$\underline{\sum F_z = 0} : 0.62 T_{AB} - 0.7 T_{AC} = 0$$

mode $\rightarrow 5 \rightarrow 2$

P	T_{AB}	T_{AC}	
1	-0.093	-0.084	0
0	0.77	0.7	1962
0	0.62	-0.7	0

$$P = 247.5$$

$$T_{AB} = 1411.5$$

$$T_{AC} = 1250.1$$

A crate is supported by three cables as shown. Determine the weight of the crate knowing that the tension in cable AB is 750 lb.

$$\vec{W} = ??$$

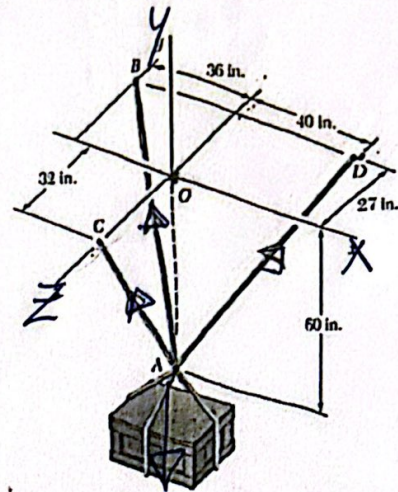
$$T_{AB} = 750 \text{ lb.}$$

$$\vec{W} = -(w)\hat{j}$$

$$\begin{aligned} \vec{T}_{AB} &= (750) \left[\frac{-36\hat{i} + 60\hat{j} - 27\hat{k}}{\sqrt{36^2 + 60^2 + 27^2}} \right] \\ &= -360\hat{i} + 600\hat{j} - 270\hat{k} \end{aligned}$$

$$\begin{aligned} \vec{T}_{AC} &= (T_{AC}) \left[\frac{0\hat{i} + 60\hat{j} + 32\hat{k}}{\sqrt{0^2 + 60^2 + 32^2}} \right] \\ &= (T_{AC}) \left[0\hat{i} + \frac{15}{17}\hat{j} + \frac{8}{17}\hat{k} \right] \end{aligned}$$

$$\begin{aligned} \vec{T}_{AD} &= (T_{AD}) \left[\frac{40\hat{i} + 60\hat{j} - 27\hat{k}}{\sqrt{40^2 + 60^2 + 27^2}} \right] \\ &= (T_{AD}) \left[\frac{40}{77}\hat{i} + \frac{60}{77}\hat{j} - \frac{27}{77}\hat{k} \right] \end{aligned}$$



$$A(0, -60, 0)$$

$$B(-36, 0, -27)$$

$$C(0, 0, 32)$$

$$D(40, 0, -27)$$

$$T_{AD} = 693 \text{ N}$$

$$T_{AC} = 1090.1 \text{ N}$$

$$W = 2101 \text{ N}$$

$$\sum F_x = 0 \rightarrow -360 + \frac{40}{77} T_{AD} = 0$$

$$\sum F_y = 0 \rightarrow -w + 600 + \frac{15}{17} T_{AC} + \frac{60}{77} T_{AD} = 0$$

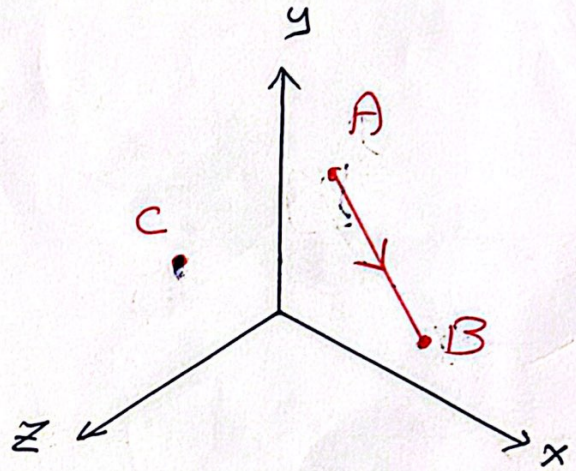
$$\sum F_z = 0 \rightarrow -270 + \frac{8}{17} T_{AC} - \frac{27}{77} T_{AD} = 0$$

* Moment in (3D) ~ (العزم)

① Moment about Point (العزم حول نقطة)

$$M_c = \vec{r} \times \vec{F}_{AB}$$

نقطة moment
أى نقطة force



We can use $\vec{r}_{CA} = A - C$

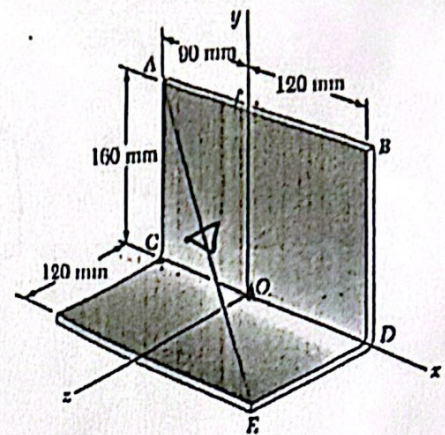
or $\vec{r}_{CB} = B - C$

$$\therefore M_c = \vec{r} \times \vec{F}_{AB}$$

$$= \begin{bmatrix} \hat{i} & \hat{j} & \hat{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{bmatrix}$$

$$= \hat{i} [r_y F_z - r_z F_y] - \hat{j} [r_x F_z - r_z F_x] + \hat{k} [r_x F_y - r_y F_x]$$

The wire AE is stretched between corners A and E of a bent plate. Knowing that the tension in the wire is 435N. Determine the moment about O of the force exerted by the wire on corner A.



$$\vec{M}_O = \vec{r}_{OA} \times \vec{T}_{AE}$$

$$\vec{r}_{OA} = A - O = -90\hat{i} + 160\hat{j} + 0\hat{k}$$

$$\vec{T}_{AE} = (435) \left(\frac{210\hat{i} - 160\hat{j} + 120\hat{k}}{\sqrt{210^2 + 160^2 + 120^2}} \right)$$

$$= 315\hat{i} - 240\hat{j} + 180\hat{k}$$

$$\vec{M} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -90 & 160 & 0 \\ 315 & -240 & 180 \end{vmatrix} = \hat{i}(28800) - \hat{j}(-16200) + \hat{k}(-28800)$$

$$M = \sqrt{28800^2 + 16200^2 + 28800^2} = 43832.86$$

Ans $M = \lambda_{\underline{OA}} \left[\underline{r}_{OB} \times \underline{T}_{BD} \right]$

• $\lambda_{OA} = \left[\frac{0\hat{i} + 2\hat{j} + 0\hat{k}}{\sqrt{0^2 + 2^2 + 0^2}} \right] = 0\hat{i} + 1\hat{j} + 0\hat{k}$

• $\underline{r}_{OB} = 2.5\hat{i} + 2\hat{j} + 0\hat{k}$

• $\underline{T}_{BD} = -300\hat{i} - 600\hat{j} + 600\hat{k}$

$M = \begin{bmatrix} \textcircled{+} & \textcircled{-} & \textcircled{+} \\ 0 & 1 & 0 \\ 2.5 & 2 & 0 \\ -300 & -600 & 600 \end{bmatrix}$

$= \cancel{-0} \begin{vmatrix} & & \end{vmatrix} - 1 \begin{vmatrix} & & \\ & 1500 & \end{vmatrix} + 0 \begin{vmatrix} & & \end{vmatrix}$

$= \boxed{-1500}$

A 2.6 lb force is applied at Point D of the cast iron post shown in Figure 2. Determine the moment of that force about the center A of the base section.

$$T_{DE} = 2.6 \text{ lb}$$

$$M_A = r_{AD} \times T_{DE}$$

$$r_{AD} = D - A = 6i + 12j + 0k$$

$$T_{DE} = (2.6) \left(\frac{0i - 12j - 5k}{\sqrt{0^2 + 12^2 + 5^2}} \right)$$

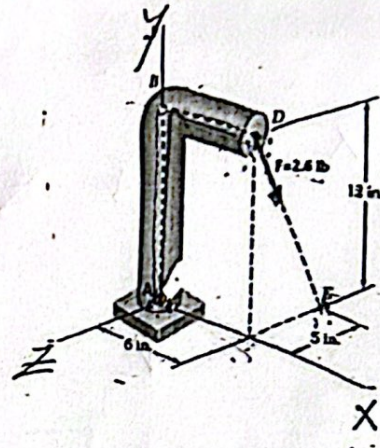
$$= 0i - 2.4j - 1k$$

$$M = \begin{bmatrix} \begin{matrix} \oplus & & \oplus \\ i & j & k \end{matrix} \\ 6 & 12 & 0 \\ 0 & -2.4 & -1 \end{bmatrix}$$

$$= i(-12) - j(-6) + k(-14.4)$$

$$M = \sqrt{12^2 + 6^2 + 14.4^2}$$

$$= 19.68 \text{ lb}\cdot\text{in}$$



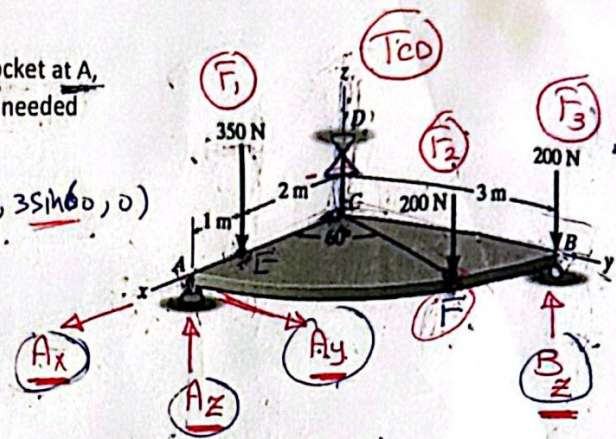
$$A(0,0,0)$$

$$D(6,12,0)$$

$$E(6,0,-5)$$

Determine the force components acting on the ball-and-socket at A, the reaction at the roller B and the tension on the cord CD needed for equilibrium of the quarter circular plate.

$A(3, 0, 0)$ $C(0, 0, 0)$ $F(3\cos 60^\circ, 3\sin 60^\circ, 0)$
 $B(0, 3, 0)$ $E(2, 0, 0)$



$\vec{A} = (A_x)\hat{i} + (A_y)\hat{j} + (A_z)\hat{k}$

$\vec{B} = (B_z)\hat{k}$

$\vec{F}_1 = -350\hat{k}$

$\vec{F}_2 = -200\hat{k}$

$\vec{F}_3 = -200\hat{k}$

$\vec{T}_{CD} = (T_{CD})\hat{k}$

$\sum \vec{F}_x = 0 : A_x = 0$

$\sum \vec{F}_y = 0 : A_y = 0$

$\sum \vec{F}_z = 0 : A_z + B_z - 200 - 200 - 350 = 0 \quad \Rightarrow \quad A_z + B_z + T_{CD} = 750 \quad \text{--- (1)}$

$\sum M_A = 0 : \sum M_A = \vec{r}_{AB} \times \vec{B}_z + \vec{r}_{AC} \times \vec{T}_{CD} + \vec{r}_{AE} \times \vec{F}_1 + \vec{r}_{AF} \times \vec{F}_2 + \vec{r}_{AB} \times \vec{F}_3 = 0$

$$= \begin{bmatrix} + & - & + \\ \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 0 & 0 & B_z \end{bmatrix} + \begin{bmatrix} + & - & + \\ \hat{i} & \hat{j} & \hat{k} \\ -3 & 0 & 0 \\ 0 & 0 & T_{CD} \end{bmatrix} + \begin{bmatrix} + & - & + \\ \hat{i} & \hat{j} & \hat{k} \\ -1 & 0 & 0 \\ 0 & 0 & -350 \end{bmatrix} + \begin{bmatrix} + & - & + \\ \hat{i} & \hat{j} & \hat{k} \\ -3 & 3 & 0 \\ 0 & 0 & -200 \end{bmatrix} + \begin{bmatrix} + & - & + \\ \hat{i} & \hat{j} & \hat{k} \\ -1.5 & 2.59 & 0 \\ 0 & 0 & -200 \end{bmatrix}$$

$\hat{i} = 0, \quad 3B_z - 600 - 518 = 0 \Rightarrow B_z = 372.6 \text{ N}$

$\hat{j} = 0, \quad 3B_z + 3T_{CD} - 350 - 600 - 300 = 0 \Rightarrow T_{CD} = 44 \text{ N}$

from eq. (1), $\therefore A_z = 333.4 \text{ N}$

(x , y , z)

$$B(0, 12, 8)$$

$$C(0, 12, -10)$$

$$A(1.2, 2, 0)$$

$$\vec{W} = -1962 \hat{j}$$

$$\vec{F} = (F) \hat{i}$$

$$\vec{T}_{AB} = (T_{AB}) \left(\frac{-1.2\hat{i} + 10\hat{j} + 8\hat{k}}{\sqrt{1.2^2 + 10^2 + 8^2}} \right) \quad \begin{matrix} \swarrow \\ B-A \end{matrix}$$

$$= (T_{AB}) (-0.0932\hat{i} + 0.77\hat{j} + 0.62\hat{k})$$

$$\vec{T}_{AC} = (T_{AC}) \left(\frac{-1.2\hat{i} + 10\hat{j} - 10\hat{k}}{\sqrt{1.2^2 + 10^2 + 10^2}} \right) \quad \begin{matrix} \nearrow \\ C-A \end{matrix}$$
$$= (T_{AC}) (-0.084\hat{i} + 0.7\hat{j} - 0.7\hat{k})$$

1- Determine the reactions at the roller support A, the ball-and-socket joint D, and the tension in cable BC for the plate.

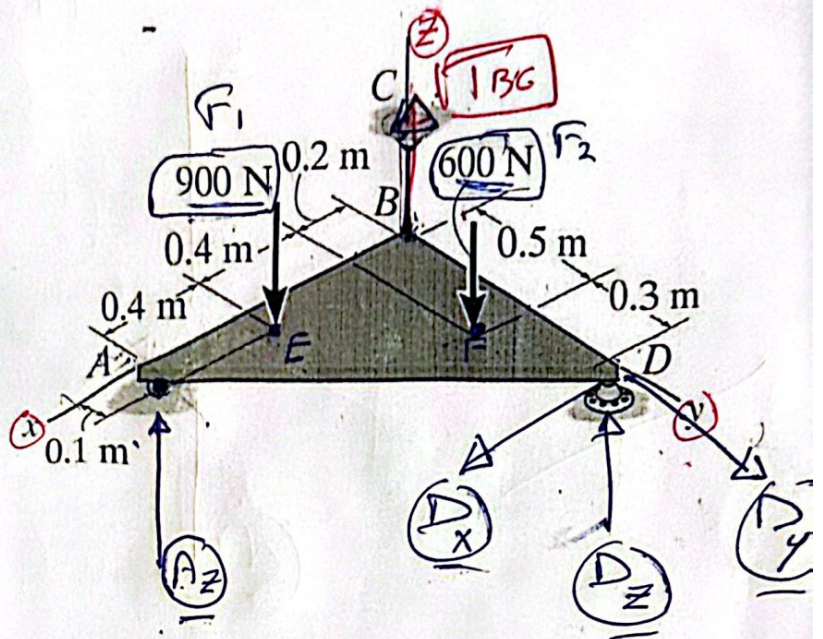
$$B (0, 0, 0)$$

$$A (1, 0.1, 0)$$

$$D (0, 0.8, 0)$$

$$E (0.6, 0.1, 0)$$

$$F (0.2, 0.5, 0)$$



$$\vec{R}_A = (A_z) \mathbf{k}$$

$$\vec{R}_D = (D_x) \mathbf{i} + (D_y) \mathbf{j} + (D_z) \mathbf{k}$$

$$\vec{T}_{BC} = (T_{BC}) \mathbf{k}$$

$$\vec{F}_1 = -900 \mathbf{k}$$

$$\vec{F}_2 = -600 \mathbf{k}$$

$$\sum F_x = 0 : \boxed{D_x = 0} \checkmark$$

$$\sum F_y = 0 : \boxed{D_y = 0} \checkmark$$

$$\sum F_z = 0 : \boxed{A_z + D_z + T_{BC} - 900 - 600 = 0} \checkmark$$

$$M_D = \underbrace{r_{DA} \times A_z}_{\text{DA}} + \underbrace{r_{DB} \times T_{BC}}_{\text{DB}} + \underbrace{r_{DE} \times F_1}_{\text{DE}} + \underbrace{r_{DF} \times F_2}_{\text{DF}}$$

$$\left. \begin{array}{l} r_{DA} = A - D = 1i - 0.7j \\ r_{DB} = B - D = -0.8j \end{array} \right\} \left. \begin{array}{l} r_{DE} = E - D = 0.6i - 0.7j \\ r_{DF} = F - D = 0.2i - 0.3j \end{array} \right\}$$

$$M_D = \begin{bmatrix} \oplus & \ominus & \oplus \\ i & j & k \\ 1 & -0.7 & 0 \\ 0 & 0 & A_z \end{bmatrix} + \begin{bmatrix} \oplus & \ominus & \oplus \\ i & j & k \\ 0 & -0.8 & 0 \\ 0 & 0 & T_{BC} \end{bmatrix} + \begin{bmatrix} \oplus & \ominus & \oplus \\ i & j & k \\ 0.6 & -0.7 & 0 \\ 0 & 0 & -900 \end{bmatrix} + \begin{bmatrix} \oplus & \ominus & \oplus \\ i & j & k \\ 0.2 & -0.3 & 0 \\ 0 & 0 & -600 \end{bmatrix}$$

$$\underline{\Sigma M_x = 0}$$

$$-0.7A_z - 0.8T_{BC} + 630 + 180 = 0$$

$$\underline{\Sigma M_y = 0}$$

$$-A_z + 540 + 120 = 0$$

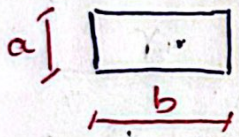
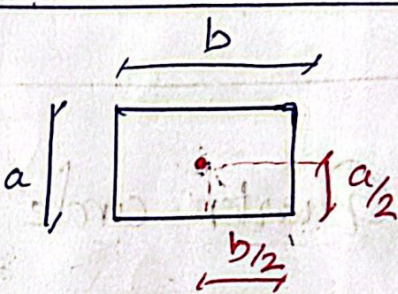
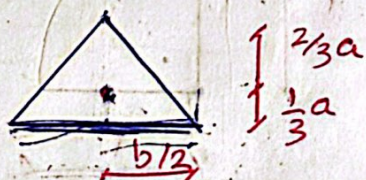
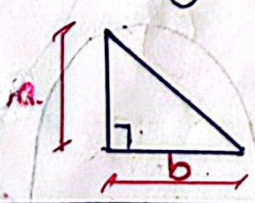
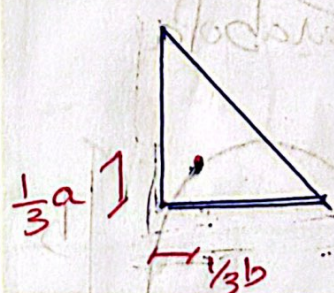
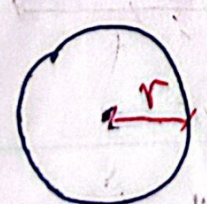
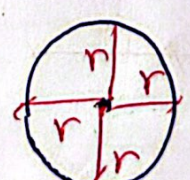
$$\underline{\Sigma M_z = 0}$$

$$\boxed{A_z = 660}$$

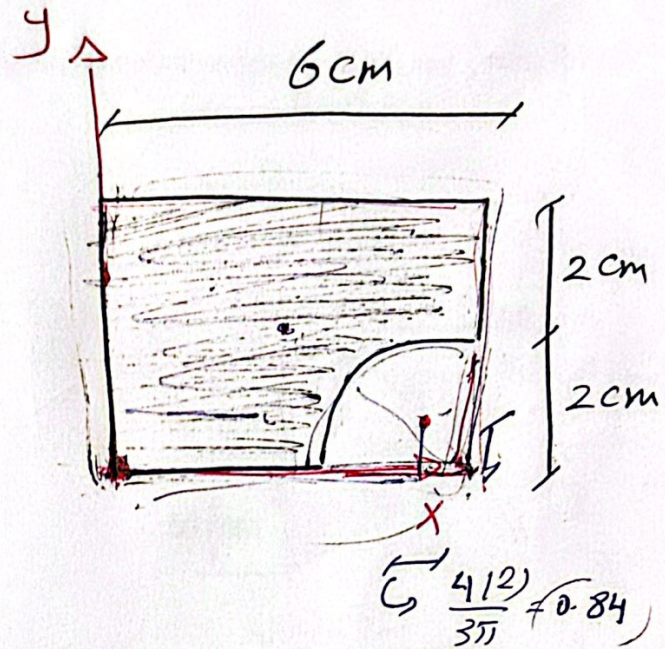
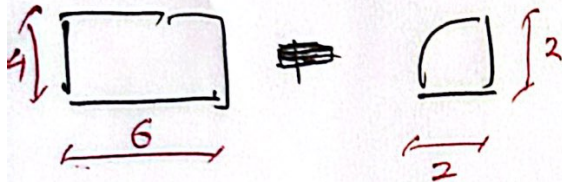
$$\boxed{T_{BC} = 435}$$

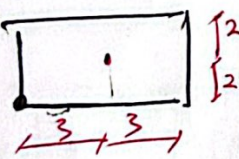
$$\boxed{D_2 = 405}$$

Centroid

Shapes	Area	Center $(\bar{x}, \bar{y})$
① Rectangle 	$a \times b$	
② Triangle 	$\frac{1}{2} \times b \times a$	
③ Right-angle Triangle 	$\frac{1}{2} \times b \times a$	
④ Circle 	πr^2	

Locate the
Centroid



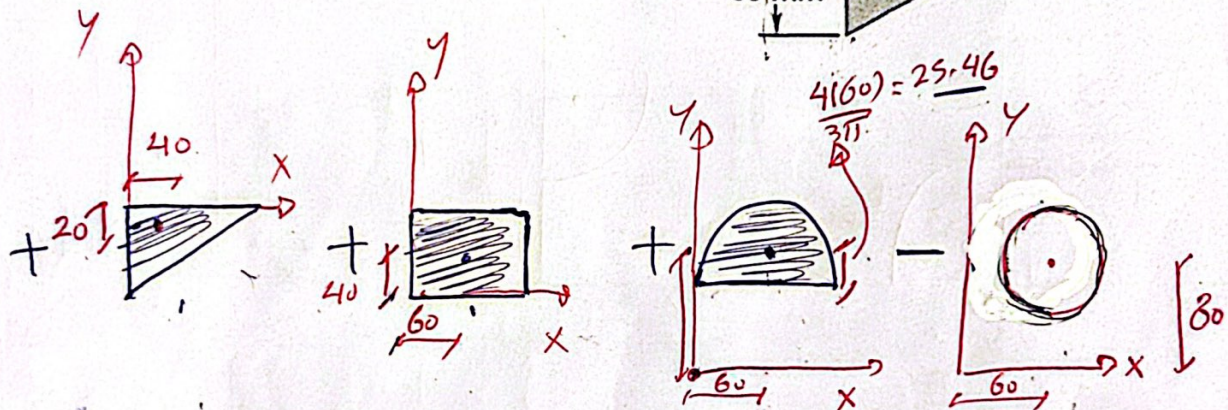
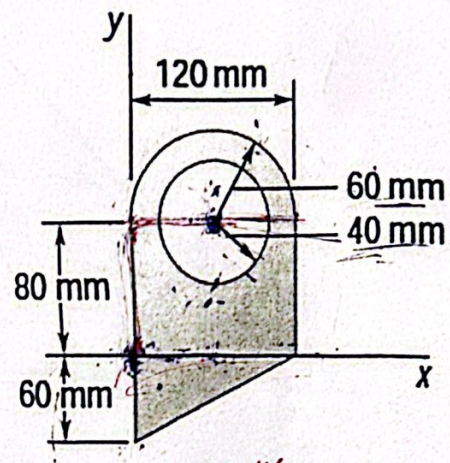
Shapes	Area	$\bar{x}$	$\bar{y}$	$\bar{x} \cdot \text{Area}$	$\bar{y} \cdot \text{Area}$
	$6 \times 4 = 24$	3	2	72	48
	$-\frac{1}{4}\pi(2)^2 = -3.14$	$6 - 0.84 = 5.16$	0.84	-16.2	-2.63
	<u>20.86</u>			<u>55.8</u>	<u>45.37</u>

$$\frac{4(2)}{3\pi} = 0.84$$

$$\bar{x} = \frac{\sum \bar{x} \cdot \text{Area}}{\sum \text{Area}} = \frac{55.8}{20.86} = \underline{\underline{2.53}}$$

$$\bar{y} = \frac{\sum \bar{y} \cdot \text{Area}}{\sum \text{Area}} = \frac{45.37}{20.86} = \underline{\underline{2.17}}$$

For the plane area shown, determine (a) the first moments with respect to the x and y axes, (b) the location of the centroid



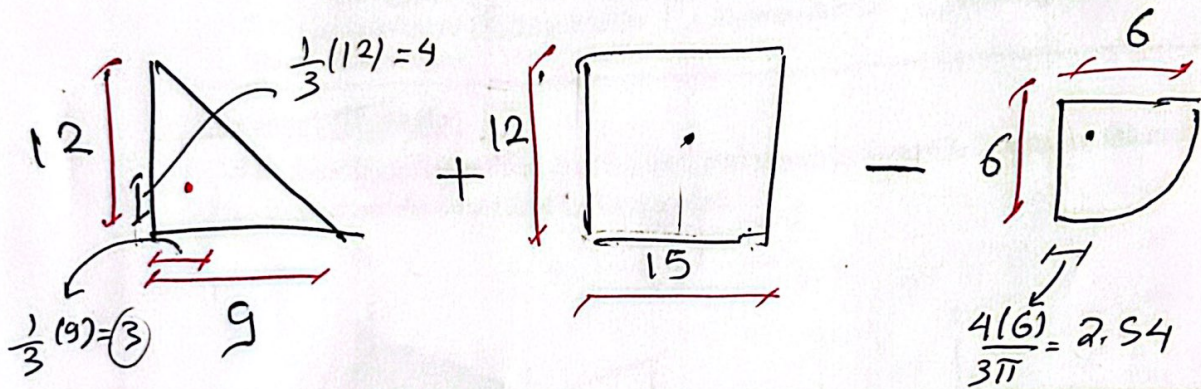
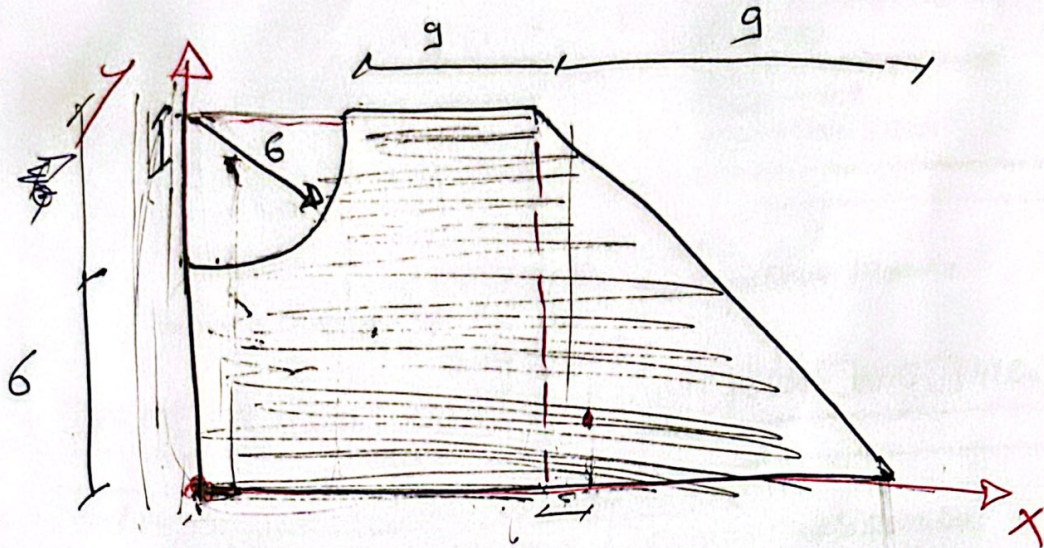
Shape	Area	$\bar{x}$	$\bar{y}$	$\bar{x} \text{ Area}$	$\bar{y} \text{ Area}$
	3600	40	-20	144000	-72000
	9600	60	40	576000	384000
	1800π	60	105.46	108000π	189828π
	-1600π	60	80	-96000π	-128000π
Σ	13828.31		Σ	757699.11	1310486.44 506238.39

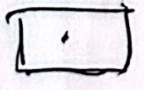
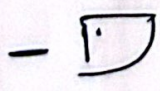
$$\bar{X} = \underline{\underline{54.79 \text{ mm}}}$$

$$\bar{Y} = \underline{\underline{36.6 \text{ mm}}}$$

(7)

Locate the centroid.



Shapes	Area	$\bar{x}$	$\bar{y}$	$x \text{ Area}$	$y \text{ Area}$
	$\frac{1}{2} \times 9 \times 12$ $= 54$	$15 + 3$ $= 18$	4	972	216
	12×6 $= 72$	6	3	432	216
	$-\frac{1}{4} \pi (6)^2$ $= -28.2$	2.54	$12 - 2.54$ $= 9.46$	-71.62	-266.77
	<u>205.8</u>			<u>2250.38</u>	<u>1029.23</u>

$$\bar{X} = \frac{2250.38}{205.8} = 10.93$$

$$\bar{Y} = \frac{1029.23}{205.8} = 5$$



Academic year: 2024/2025
Semester: 1st semester
Exam: Final Term Exam
Code: BAS 031
Date: 11/1/2025

Faculty of Engineering and
Technology
Basic Science Department
Course level: 0
Allowed Time: 3 Hour

Course Title: Mechanics (1) Statics

Number of Exam Questions: 4

Max Exam Grade: 50 marks

Course Offered By

☒ FRE ☐ STR ☐ ART ☐ ENR ☐ MET ☐ CMP ☐ COM ☐ ENG ☐ HUM

Exam Instructions (plagiarized answers will receive zero marks)

1. Type of Exam: ☒ Closed-Book ☐ Open-Book ☐ Charts and Formula Sheets Only
2. Write your answers in: ☒ Same Booklet ☐ Separate Booklet ☐ Drawing Sheet
3. Attempt ALL questions

Question-1 (10 marks)

A 2.6 lb force is applied at Point D of the bent post shown in Figure 1. Determine the moment of that force about the center A of the base section.

Figure 1

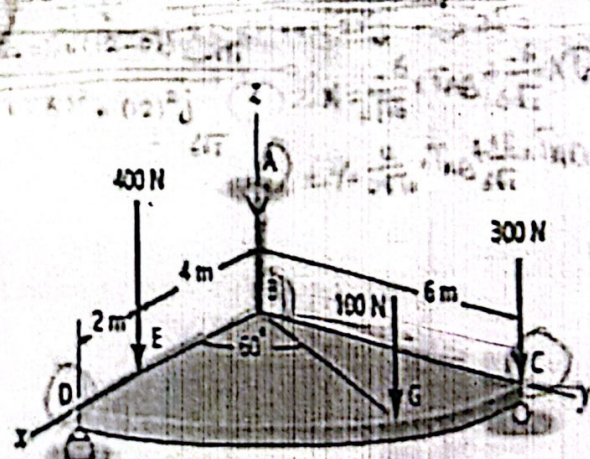
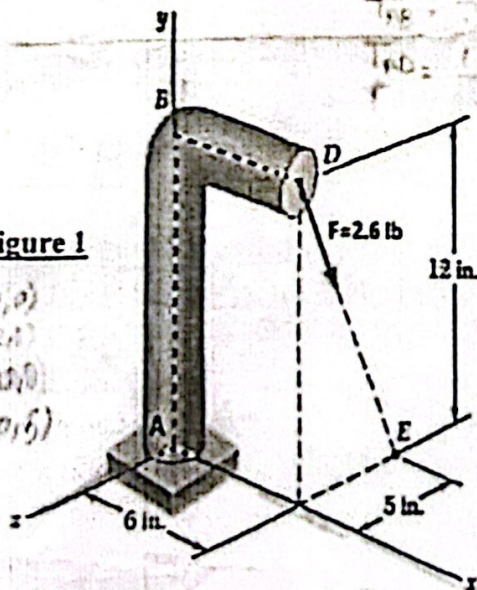


Figure 2

Question-2 (15 marks)

Determine the force components acting on the ball-and-socket at D, the reaction at the roller C and the tension on the cord AB needed for equilibrium of the quarter circular plate shown in Figure (2). Neglect the mass of the plate.



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Basic Science Department
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Question-3 (12 marks)

For the truss and load shown in Figure (3), Determine;

- 1) the reactions at rocker support E and hinge support A,
- 2) the forces in members BC, CE, and ED, and indicate whether the members are in tension or compression using the method of sections.

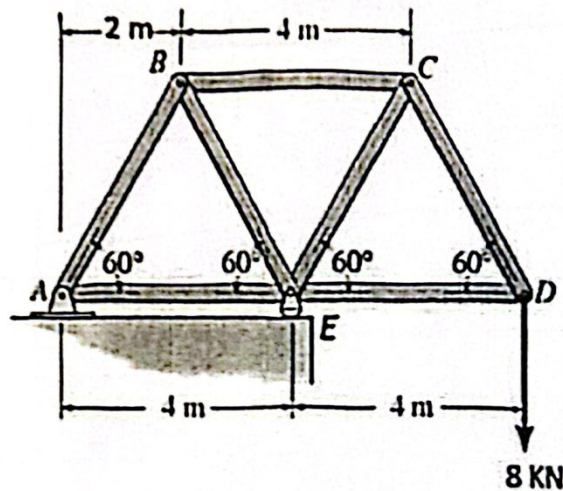


Figure 3

Question-4 (13 marks)

For the plane area shown in Figure (4), determine the location of the centroid.

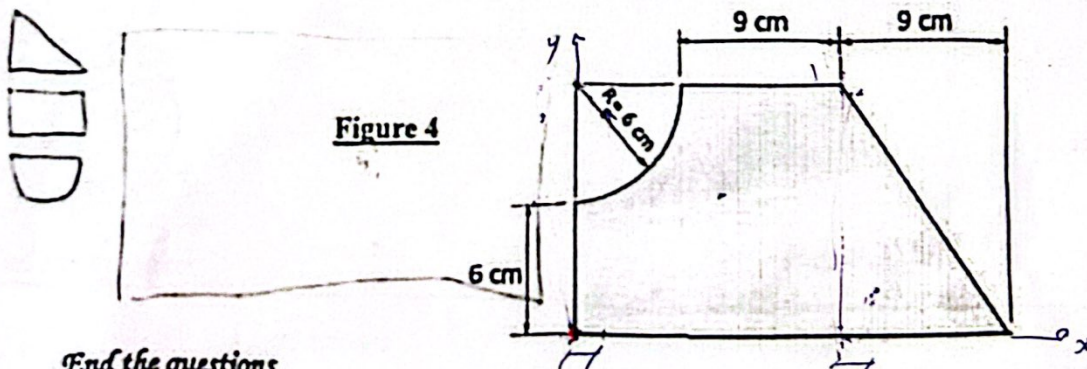


Figure 4

End the questions

With best wishes

Dr. Ahmed El-Desouky



Question-2 (15 marks)

Determine the force components acting on the ball-and-socket at D, the reaction at the roller C and the tension on the cord AB needed for equilibrium of the quarter circular plate shown in Figure (2). Neglect the mass of the plate.

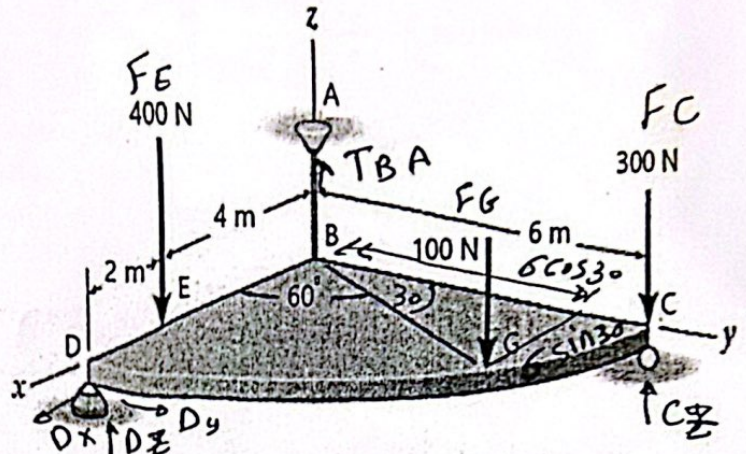
$$\vec{D} = D_x \vec{i} + D_y \vec{j} + D_z \vec{k}$$

$$\vec{C} = 0 \vec{i} + 0 \vec{j} + C_z \vec{k}$$

$$\vec{T}_{BA} = T_{BA} \vec{k}$$

$$\vec{F}_E = -400 \vec{k} \quad \vec{F}_G = -100 \vec{k}$$

$$\vec{F}_C = -300 \vec{k}$$



$$\sum F_x = 0 \quad D_x = 0$$

$$\sum F_y = 0 \quad D_y = 0$$

$$\sum F_z = 0 \quad D_z + C_z + T_{BA} - 800 = 0$$

$$\sum M_B = 0 \quad (\vec{r}_{BD} \times \vec{D}) + (\vec{r}_{BE} \times \vec{F}_E) + (\vec{r}_{BC} \times \vec{F}_C) + (\vec{r}_{BG} \times \vec{F}_G) = 0$$

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 & 0 & 0 \\ 0 & 0 & D_z \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 4 & 0 & 0 \\ 0 & 0 & -400 \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 6 & 0 \\ 0 & 0 & -300 \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 6 \cos 30^\circ & 6 \sin 30^\circ & 0 \\ 0 & 0 & -100 \end{vmatrix} = 0$$

$$\sum M_x = 0 \quad -300 \times 6 + 6 C_z - 600 \cos 30^\circ = 0$$

$$\sum M_y = 0 \quad 6 D_z - 1600 - 600 \sin 30^\circ = 0$$

$$C_z = 386.6 \text{ N}$$

$$D_z = 316.67 \text{ N}$$

$$\therefore T_{BA} = 96.73 \text{ N}$$